Hands-on Labs

Morning Session

Day 4 - Morning Labs

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Lab 4.1: Add a Moderation Layer

Security & Responsible Design

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Lab 4.1: Input/Output Moderation Layer

Goal: Add security guardrails to the Day 3 RAG chatbot.

Requirements:

1. Add input filtering: detect and block prompt injection attempts
2. Add PII redaction: strip email, phone, SSN from user input
3. Add output filtering: block responses containing forbidden content
4. Log all blocked attempts with timestamps
5. Test with at least 3 injection attempts and 3 normal queries

Time: 30 minutes | File: `gen-ai-companion/day4/morning-labs/start/lab4_1_moderation_layer.py`

Lab 4.2: Caching & Cost Measurement

Cost & Latency

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Lab 4.2: Add Caching & Measure Savings

Goal: Add caching to the RAG chatbot and measure token/cost savings.

Requirements:

1. Implement an in-memory response cache with TTL
2. Track cache hits vs misses
3. Count tokens for every API call (input + output)
4. Calculate and display cost savings from caching
5. Add a `/stats` command showing usage and savings

Lab 4.2: Add Caching & Measure Savings

Stretch goals:

- Implement model tiering (simple queries use mini, complex use default)
- Add a budget guard that switches to mini when daily budget is 80% spent

Time: 30 minutes | File: `gen-ai-companion/day4/morning-labs/start/lab4_2_caching_cost.py`